

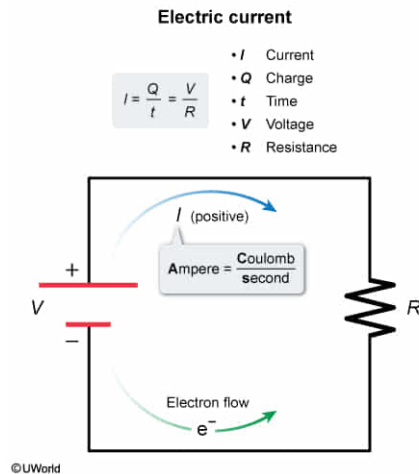
Which unit ratio CANNOT be used to express the quantity of current traveling through a circuit?

- ☐ A. $\text{W} \cdot \text{V}^{-1}$ [20%]
☐ B. $\text{C} \cdot \text{s}^{-1}$ [5%]
☒ C. $\text{F} \cdot \text{m}^{-1}$ [68%]
☐ D. $\text{V} \cdot \Omega^{-1}$ [5%]

Correct

68% answered correctly

Explanation:



Current refers to the movement of electric charge. The application of an electric field to **conductive materials** results in the concerted movement of many charged particles (electric current) in a direction aligned with the **electric field**.

If one adopts the same convention used for **electric fields**, the direction of current flow is taken to point from positive to negative charge. However, for most states of matter the only charged particles that travel throughout a material are negatively charged electrons. As a result, the direction of current flow (by convention) and the actual direction of electron flow are typically opposite one another.

In mathematical terms, current (I) is defined as the quantity of charge (Q) traveling through an object per unit time (t). Consequently, the SI unit for current is the ampere (A), where 1 A is equal to 1 coulomb (C) of charge moved per 1 second (s) of time (**Choice B**).

In practical applications, current is often produced by first establishing a voltage gradient (or electromotive force) through the use of a battery or generator. Ohm's law relates the voltage (V) generated by such a device to the total resistance (R) of the object through which the current travels:

$$I = \frac{V}{R}$$

The quantity of current traveling through a simple circuit, in combination with voltage, will influence the expenditure of power (P) for that device:

$$P = I V$$

Therefore, the current of a simple circuit or similar device may be represented in terms of volts (V) per ohm (Ω) or watts (W) per volt (V), both of which are unit ratios based on **SI units** for standard electrical variables (**Choices A and D**).

The permittivity (ϵ) of a material is commonly stated in farads (F) per meter (m), which influences the **dielectric constant** (k) of the material. Although permittivity may influence the ease at which a dipole may be induced within a substance, $\text{F} \cdot \text{m}^{-1}$ cannot be used to express the quantity of current traveling through a circuit.

Educational objective:

Current is defined as the movement of charge per unit of time. The current within a circuit may be expressed in terms of coulombs per second, volts per ohm, or watts per volt.